

Neo Li

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Education

University of Toronto

Toronto, ON

Bachelor of Applied Science in Computer Engineering + PEY Co-op

Expected April 2028

- Minor in Artificial Intelligence and Engineering Business
- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, System Design

Experience

Software Engineer Intern

May 2026 - Aug 2026

Microchip Technology

Ottawa, ON

- Engaged in full lifecycle product development within an Agile/Scrum team, applying modern design patterns to modernize a critical enterprise debugging tool
- Ported a legacy Perl codebase to a unified **Python** and **C++** architecture, actively making code changes that directly improved system interoperability and fulfilled cross-platform Linux requirements
- Engineered a CI/CD pipeline using Jenkins and Docker to streamline multi-platform builds, reducing manual deployment time by 40% and reinforcing continuous integration best practices
- Authored 400+ automated tests with pytest and resolved 45+ software bugs, consistently shipping maintainable, high-quality code across a 23K-line codebase

Director of Technology

Oct 2025 - Present

UofT Open-Source Intelligence

Toronto, ON

- Led a cross-functional team of 10+ engineers to architect an **Artificial Intelligence** platform, exploring **agentic AI** methodologies and machine learning frameworks to parse synthetic media with 80% accuracy
- Researched and integrated the **Model Context Protocol (MCP)** to standardize contextual data feeding between localized LLMs and our open-source intelligence databases
- Deployed scalable backend infrastructure using Docker and Azure Kubernetes Service (AKS), ensuring highly available, real-time data processing for student researchers

Projects

GameLink — Voice & Text Chat for Gamers | *JavaScript, Node.js, Express, Socket.IO, Docker, Jest* | [Link](#)

- Architected full-stack application with JWT auth, bcrypt hashing, and SQLite, implementing rate-limited REST APIs serving 100+ req/min for secure user authentication
- Delivered real-time bidirectional communication via Socket.IO with online presence tracking and sub-second latency across 50+ concurrent WebSocket connections for active gaming communities
- Drove end-to-end DevOps automation by containerizing the application with Docker, orchestrating multi-service deployments via docker-compose, and establishing a zero-downtime CI/CD pipeline on Render and Vercel
- Built responsive dark-mode frontend with markdown rendering, search, admin dashboard, and 100% Jest test coverage on all authentication routes ensuring reliable production-ready code

Azure MLOps Predictive Maintenance System | *Python, Azure ML, Scikit-learn, Docker* | [Link](#)

- Architected a cloud-native ML solution utilizing **Python** and Scikit-learn, designing data pipelines to train and deploy a Random Forest regression model for real-time telemetry inference
- Containerized the machine learning framework using Docker and deployed the artifact to an Azure Managed Online Endpoint, establishing a scalable architecture with zero-downtime availability

Financial Data Analyzer | *Python, Flask, React* | [Link](#) | [Demo](#)

- Engineered a full-stack tool to analyze capital market sentiment, featuring a **Python/Flask** REST API and a React frontend that processes live social data scraped from Reddit

Technical Skills

Languages: Java, Python, C, C++, JavaScript, TypeScript, SQL, HTML/CSS

AI & Machine Learning: Agentic AI, Model Context Protocol (MCP), Scikit-learn, PyTorch, Azure ML

Frameworks & Concepts: Object-Oriented Programming (OOP), Data Structures, Algorithms, System Design Patterns, Full Lifecycle Product Development, Agile Methodologies, RESTful APIs, CI/CD

Developer Tools: Git/GitHub, Docker, Kubernetes, Azure DevOps, Linux, CMake